

Project Design Phase-II

Data Flow Diagram & User Stories

Date	16 February 2026
Team ID	LTVIP2026TMIDS63806
Project Name	Heart Disease Analysis
Maximum Marks	4 Marks

Data Flow Diagrams:

A **Data Flow Diagram (DFD)** visually represents how patient health data moves through the Heart Disease Analysis system. It illustrates:

- How medical data enters the system (CSV files, hospital EMR exports, databases)
- How the data is cleaned and processed
- How risk analysis and calculations are performed
- Where the processed data is stored
- How insights are displayed through dashboards and reports

Basic Data Flow (Conceptual Overview):

Data Sources → Data Cleaning & Processing → Risk Analysis Engine → Dashboard Visualization → Reports/Alerts

Data Sources:

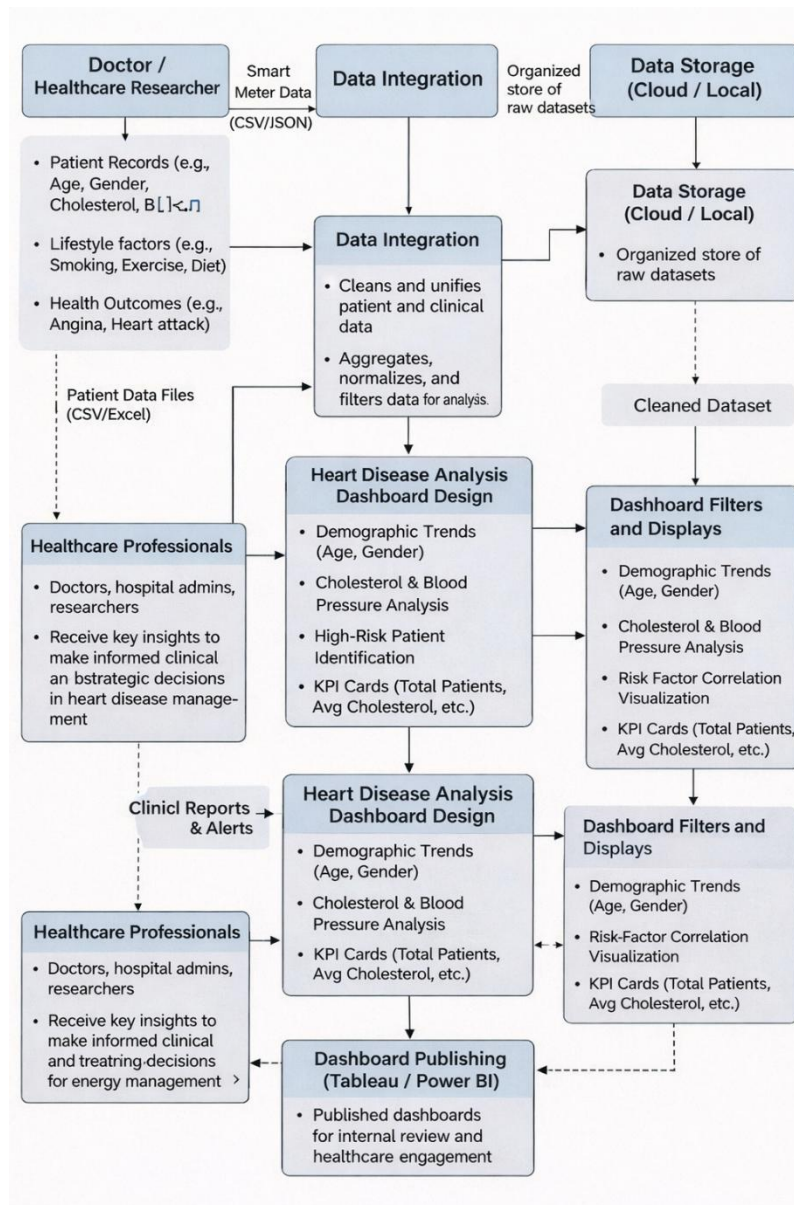
- Patient records (Age, Gender)
- Clinical metrics (Cholesterol, Blood Pressure, Heart Rate)
- Lifestyle indicators (Smoking, Exercise, etc.)

Processes:

- Data cleaning & formatting
- Risk categorization (High/Medium/Low)
- Correlation analysis
- KPI calculation

Outputs:

- Interactive dashboard
- Risk summary reports
- High-risk patient alerts



User Stories:

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Cardiologist	Data Access & Integration	USN-1	As a user, I can import heart disease patient data from CSV/Excel files	Data loads successfully and is available for analysis	High	Sprint-1
Data Analyst	Data Cleaning	USN-2	As a user, I can clean and standardize patient demographic and clinical data	Dataset is structured, formatted, and free from inconsistencies	High	Sprint-1
Cardiologist	Dashboard Overview	USN-3	As a user, I can view total patients, high-risk %, and key KPIs	KPI cards display totals, average cholesterol, BP levels, and risk %	High	Sprint-1
Hospital Administrator	Demographic Analysis	USN-4	As a user, I can compare heart disease cases by age group and gender	Visualizations update accurately for demographic comparisons	High	Sprint-1
Public Health Researcher	Risk Factor Analysis	USN-5	As a user, I can analyze relationships between cholesterol, blood pressure, and heart disease	Charts correctly reflect correlations between risk factors	High	Sprint-1
Medical Officer	Filters & Interaction	USN-6	As a user, I can filter data by age, gender, cholesterol range, and BP levels	Dashboard updates dynamically based on selected filters	High	Sprint-1
Cardiologist	High-Risk Identification	USN-7	As a user, I can identify high-risk patient groups	High-risk segments are clearly highlighted in the dashboard	Medium	Sprint-2
Healthcare Planner	Insight Generation	USN-8	As a user, I can identify key contributing factors to heart disease	Major risk factors are visually distinguishable	Medium	Sprint-2
Senior Healthcare Executive	Story & Reporting	USN-9	As a user, I can view story-driven insights for presentations	Dashboard includes structured insights with explanations	Medium	Sprint-2

System Administrator	Dashboard Publishing	USN-10	As an admin, I can publish dashboards to Tableau Server/Public	Dashboard is securely accessible to authorized stakeholders	Medium	Sprint-2
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